WEEK 5 MAIN ARTIFACT

Current-State → Future-State Process Map

Assignment 2 — Human–AI Operating Model for Company B

Student / Group:	Kniazev Daniil / AIB-2602	**Company B:**	Magnum Cash&Carry
Selected process:	Replenishment of one product category in one Magnum store (the Week 4 process)	**Week 3 AI opportunity:**	AI demand forecasting with draft order quantities (Stock planning: PILOT)

INSTRUCTIONS: Map ONE real Company B process before and after the proposed Human–AI change.
Your map must clearly show: (1) current bottleneck, (2) future Human/AI roles, (3) KPI,
(4) human review point, (5) escalation path, and (6) manual fallback. Use the SAME process developed in Week 4.

PART A — CURRENT-STATE PROCESS: How does the process work now?

Current state is a baseline to confirm with Magnum staff: public sources establish the retail and product-acceptance context, not the internal ordering workflow [14], [22].

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
What happens?	What happens?	What happens?	What happens?	What happens?	What happens?
Read sales and stock levels for the category from store records.	Check promotions and supplier limits such as delivery days.	Estimate demand and order quantity with the existing method.	Approve the order.	Send the order; receive and accept goods.	Check shelves; record availability and expired or damaged stock.
Who does it? Human / System	Who does it? Human / System	Who does it? Human / System	Who does it? Human / System	Who does it? Human / System	Who does it? Human / System
Human + System: store staff with the stock system	Human: category manager	Human: category manager	Human: authorised manager	Human + System: purchasing and receiving staff	Human: store staff

⚠ BOTTLENECK — What slows down or weakens the current process?	CURRENT KPI / BASELINE — What can you measure now?
Steps 2–3: stock records and promotion information are reconciled by hand before quantities are set. During promotions recorded stock can differ from shelf stock (Week 3 stress test), so orders can be too large or too small, causing stockouts or waste. Working hypothesis to confirm with Magnum staff.	**KPI:** Shelf availability = checks with the product available ÷ all scheduled checks × 100. Supporting: forecast error (WAPE). **Current value (if available):** not published by Magnum. Record a baseline before the pilot, using the same products and checking schedule.

PART B — FUTURE-STATE PROCESS: How should the Human–AI process work?

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Future action	Future action	Future action	Future action	Future action	Future action
Validate sales and stock inputs: missing fields, data age.	Forecast demand and draft order quantities.	Adjust for promotions, delivery days and supplier limits.	Approve, change or reject the order.	Send the approved order; receive and accept goods.	Check shelves and costs; log inputs, revisions and approvals.
Performer Human / AI / Both	Performer Human / AI / Both	Performer Human / AI / Both	Performer Human / AI / Both	Performer Human / AI / Both	Performer Human / AI / Both
Automated fixed rules (not AI)	AI, reviewed by the category manager	Both: planner and store staff with AI flags	Human only: category manager	Both: purchasing and receiving staff with the system	Both: store staff and automated log
Control / check	Control / check	Control / check	Control / check	Control / check	Control / check
Failed or stale input blocks the recommendation and alerts staff.	Draft only; it cannot be sent. The existing method runs on the same inputs for comparison.	Each change and its reason is recorded.	Checks physical stock, sales, promotions, shelf life and budget. No release without approval.	Only approved orders can be sent. Goods are accepted under Magnum rules [22].	Manager reviews exceptions daily and trends weekly.

HUMAN REVIEW POINT Where must a human check or approve AI output?	ESCALATION PATH Where does a difficult / uncertain case go?
Step 4. The category manager checks physical stock, sales, promotions, shelf life and budget before any AI-drafted order is released, and approves, changes or rejects it. Staff may challenge a forecast at Step 3 [1], [16].	**AI** → Category manager (data or forecast issue) → Supply-chain lead (unresolved supply issue). Technical fault → IT. Repeated failures pause the trial.

MANUAL FALLBACK **What happens if AI/system is unavailable?**	**TARGET KPI** **How will you know the future process works better?**
Stop the affected AI recommendation. Staff count physical stock and use verified sales to draft a manual order; the category manager approves it and the process rejoins at Step 5. Restart only after the manager confirms the fault is corrected [16].	**KPI:** Shelf availability **Current:** to be measured (not public) **Target:** ≥ 98% Also: WAPE 10% below baseline; waste value and staff cost per order 10% lower each; net benefit positive. Proposed pilot targets, not Magnum results.

PART C — MANAGEMENT JUSTIFICATION

What improves?	Time / cost / quality / customer experience / productivity / other: Fewer empty shelves, less expired or damaged stock, less staff time per order and lower forecast error, tested against current ordering.
What remains human?	Which decision or activity still requires human judgment/accountability? The category manager approves, changes or rejects every order, authorises restart after a failure and stays accountable; staff add local knowledge.
What does AI change?	Which activities are augmented or automated? Augmented: forecasts and draft quantities (AI executes, human reviews) and promotion adjustments (human + AI). Fixed-rule automation, not AI: input checks and logs. Approval is not automated.
Main risk + control	What is the main risk, and what control reduces it? Risk: wrong stock or promotion records give confident but wrong quantities. Control: input checks, manager approval and WAPE read with shelf checks; if recorded and shelf stock differ, DELAY until reconciled (Week 3 stop condition).

FINAL ONE-SENTENCE RECOMMENDATION

Company B should PILOT AI demand forecasting with draft order quantities in one store and product category for six months, keeping the category manager's approval for every order,
because Carrefour's forecasting initiative shows the practice is worth testing [20], while Magnum's data quality, results and costs stay unproven until they are compared with current ordering.

SUBMISSION CHECK

☒ Current-state map ☒ Bottleneck ☒ Future-state map ☒ KPI ☒ Human review point ☒ Escalation path ☒ Manual fallback

Sources (cumulative Evidence Folder numbers): [1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 7.4, 10.1, 10.4–10.5. [14] Magnum Cash&Carry, "About us," n.d. [16] NIST, "Artificial Intelligence Risk Management Framework AI RMF 1.0," NIST AI 100-1, Jan. 2023. [20] SAS, "Multinational Carrefour relies on SAS AI to optimize supply chain management and reduce food waste," Jan. 14, 2019. [22] Magnum Cash&Carry, "Corporate rules," n.d.